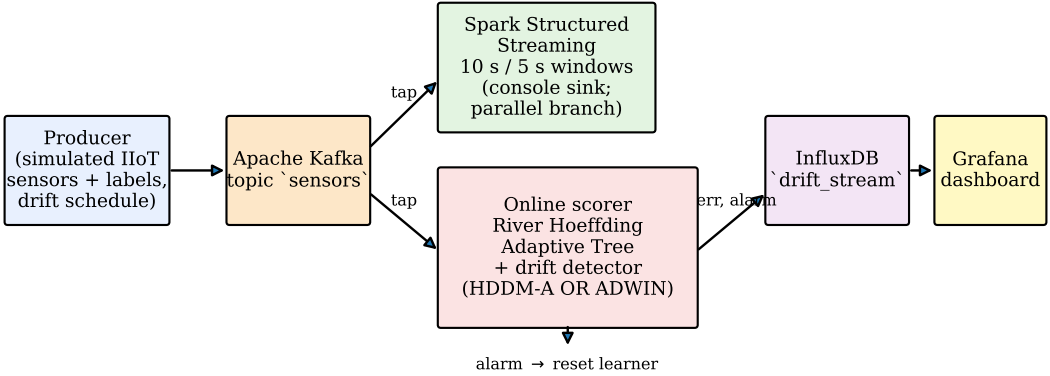
Producer
(simulated IIoT
sensors + labels,
drift schedule)
Apache Kafka
topic `sensors`
tap
tap
Spark Structured
Streaming
10 s / 5 s windows
(console sink;
parallel branch)
Online scorer
River Hoeffding
Adaptive Tree
+ drift detector
(HDDM-A OR ADWIN)
err, alarm
alarm → reset learner
InfluxDB
`drift_stream`
Grafana
dashboard